

Centrifugal Pump Seal Inspection Manual

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1. Safety Guidelines

WARNING: Before performing any inspection, ensure the pump is isolated from the power supply. Lock out the main breaker and follow your facility's LOTO (Lockout/Tagout) procedures. Always wear appropriate PPE including safety glasses, gloves, and closed-toe shoes.

Mechanical seal systems operate under high pressure and temperature. Improper handling can result in personal injury, equipment damage, or loss of containment. Ensure all personnel are trained before performing maintenance.

2. Seal Operating Parameters (Section 4.2)

2.1 Temperature Specifications

Seal operating temperature range: **20-45°C (68-113°F)**

Critical threshold: 50°C (122°F) - If exceeded, reduce pump speed or increase cooling flow.

Emergency shutdown: >60°C (140°F) - Immediately stop pump and investigate temperature rise.

2.2 Vibration Specifications

Bearing vibration limit: **< 3 m/s (peak velocity)**

Warning threshold: 2.0-2.5 m/s - Schedule maintenance within 1 month

Alert threshold: 2.5-3.0 m/s - Schedule maintenance within 1 week

Shutdown threshold: > 3.0 m/s - Stop pump immediately and diagnose

2.3 Pressure Specifications

Normal discharge pressure: **60-90 psi (4.1-6.2 bar)**

Suction pressure: 15-25 psi (1.0-1.7 bar) above atmospheric

Pressure differential: < 10 psi indicates potential seal degradation

2.4 Electrical Current

Normal motor current: 30-40 Amps (typical 1-3 HP motor)

Current spike on startup: Expected up to 50 Amps (2-3 seconds only)
If sustained high current: Check for cavitation or seal blockage

3. Inspection Procedures (Section 5.1)

Daily Visual Inspection:

- Check for leaks around the pump shaft (acceptable: light drips only)
- Listen for unusual noises (grinding, squealing indicates damage)
- Touch the pump casing (at arm's length) - should feel warm but not hot
- Log baseline vibration reading using accelerometer on pump housing

Monthly Detailed Inspection:

- Measure vibration with certified equipment at 3 points (horizontal, vertical, axial)
- Record temperature using infrared thermometer on discharge line
- Measure suction and discharge pressures
- Record motor electrical current on all 3 phases

4. Maintenance Schedule

Maintenance Task	Frequency	Criticality
Visual inspection	Daily	Low
Vibration measurement	Monthly	Medium
Temperature monitoring	Monthly	Medium
Seal inspection for leaks	500 operating hours	High
Mechanical seal replacement	Every 2 years or 8000 hrs	High
Bearing lubrication check	Every 3 months	Medium
Pressure test	Annually	Medium

5. Troubleshooting Guide

Problem: High seal temperature (>50°C)

Possible causes: Low cooling water flow, internal friction, cavitation

Actions: Check cooling water supply, reduce pump speed, verify NPSH (Net Positive Suction Head)

Problem: Abnormal vibration (>2.5 m/s)

Possible causes: Impeller imbalance, bearing wear, misalignment, cavitation

Actions: Check suction pressure, verify pump alignment, inspect bearings

Problem: Seal leakage at shaft

Possible causes: Seal wear, damaged O-rings, pressure surge

Actions: Reduce pressure gradually, replace mechanical seal within 24-48 hours

Problem: Increased power consumption

Possible causes: Cavitation, seal friction, impeller fouling

Actions: Check suction line blockage, increase inlet pressure, verify discharge line restriction